

TradingAI Milestone History - Day 1 to Current

Complete implementation history through Milestone 78

System reference date: 2026-08-29. This is the standalone milestone history for the current platform.

The history is reconstructed from the current source tree and supplied milestone/thread context. Earlier milestones are described by what they introduced and the role those capabilities play in the system that exists today.

1. How the platform evolved

TradingAI evolved from indicator/backtest foundations into a governed institutional workflow spanning market intelligence, Stock Intelligence, options strategy/valuation, Decision Intelligence, portfolio risk/allocation, execution, autonomous management, performance learning and isolated historical/prospective research. The implementation is cumulative: later milestones harden and integrate earlier layers.

Milestone 1-26 - Core platform foundation

Purpose. Created the base abstractions that later milestones reused.

Implemented capability. Repository architecture; PostgreSQL/Alembic; historical price storage; technical indicators; feature engineering; signal/scoring engines; historical backtesting; HTML reporting; experiment runner; configuration, CLI and repository/reporting frameworks.

Primary inputs. Raw historical market observations and configuration.

Primary outputs / authority. Persisted features/signals/backtest results/reports.

Current architectural role. Created the base abstractions that later milestones reused.

Milestone 27 - Historical Option Chain Foundation

Purpose. Made options research evidence-based rather than synthetic-only.

Implemented capability. Historical option-chain database/repository, pricing service, trade generator, liquidity filters, historical option selection, diagnostics/reporting/backtesting and multiple pricing/report correctness fixes.

Primary inputs. Historical underlying and option-chain observations.

Primary outputs / authority. Historically priced option trades with coverage/rejection diagnostics.

Current architectural role. Made options research evidence-based rather than synthetic-only.

Milestone 28 - Institutional Options Strategy Engine

Purpose. Established a structured options decision object consumed by later risk/decision work.

Implemented capability. Volatility intelligence; strategy, strike and expiration selection; Greeks and liquidity optimization; expected move; strategy scoring/ranking; multi-strategy support; portfolio construction; institutional decision bundle.

Primary inputs. Underlying signal/features plus option chain.

Primary outputs / authority. InstitutionalDecision, DecisionCandidateBundle, approvals, rankings, allocations, warnings and rejections.

Current architectural role. Established a structured options decision object consumed by later risk/decision work.

Milestone 29 - Institutional Probability & Risk Analytics

Purpose. Shifted the platform from score-only selection toward probability/risk-aware institutional decisioning.

Implemented capability. Probability engine, scenario analytics, distribution/tail risk, risk surfaces, portfolio optimization, probability calibration, walk-forward validation, market-regime analytics and execution analytics.

Primary inputs. Trade/strategy candidates, distributions, historical outcomes and market regime.

Primary outputs / authority. POP/EV/Kelly/confidence, VaR/ES/tail metrics, scenario approvals, calibrated probabilities and execution-quality analytics.

Current architectural role. Shifted the platform from score-only selection toward probability/risk-aware institutional decisioning.

Milestone 30 - Production-grade platform hardening

Purpose. Separated model correctness from operational readiness.

Implemented capability. Monitoring, health checks, logging, alerting, performance tuning, deployment automation, operational resilience and production-readiness foundations.

Primary inputs. Application/runtime state.

Primary outputs / authority. Operational health and deployment controls.

Current architectural role. Separated model correctness from operational readiness.

Milestone 34 - Institutional Research Workstation

Purpose. Learning was explicitly advisory; it could not mutate production authority automatically.

Implemented capability. Research knowledge cases, evidence/scenario/decision/outcome records, similarity/pattern discovery, institutional learning, analyst performance and knowledge dashboard.

Primary inputs. Research cases and prior outcomes.

Primary outputs / authority. Searchable knowledge profiles, pattern clusters, learning recommendations, analyst scorecards and readiness dashboard.

Current architectural role. Learning was explicitly advisory; it could not mutate production authority automatically.

Milestone 35 - Institutional Market Scanner

Purpose. Closed the gap between batch analytics and a daily actionable discovery workflow.

Implemented capability. Institutional scanner foundation, market data population/quality, universe handling, options-data coverage and ranking-oriented scanner outputs.

Primary inputs. Canonical universe, underlying/options observations.

Primary outputs / authority. Ranked market opportunities and scanner diagnostics.

Current architectural role. Closed the gap between batch analytics and a daily actionable discovery workflow.

Milestone 36 - Opportunity Analysis Workbench

Purpose. Introduced the candidate-centric workflow later formalized by M54/M62.

Implemented capability. Deep candidate inspection and strategy-analysis workflow, subsequently folded into the canonical opportunity and institutional-options architecture.

Primary inputs. Scanner candidate.

Primary outputs / authority. Inspectable opportunity evidence and strategy comparison.

Current architectural role. Introduced the candidate-centric workflow later formalized by M54/M62.

Milestone 37 - Institutional Decision Workspace / Risk

Purpose. Made risk an explicit downstream authority rather than a report-only afterthought.

Implemented capability. Decision presentation plus mature portfolio risk-management capabilities: Greeks, exposures, limits, scenarios, risk breaches and decision integration.

Primary inputs. Candidate plus portfolio state.

Primary outputs / authority. Institutional decision explanation and risk-aware accept/reject evidence.

Current architectural role. Made risk an explicit downstream authority rather than a report-only afterthought.

Milestone 38 - Paper Trading & Execution Orchestration

Purpose. Created the path from analysis to controlled simulated execution.

Implemented capability. Execution orchestration and paper-trading handoff across order, execution and portfolio layers.

Primary inputs. Approved trade intent.

Primary outputs / authority. Paper orders/events and execution lifecycle state.

Current architectural role. Created the path from analysis to controlled simulated execution.

Milestone 39 - Position Monitoring / Performance

Purpose. Closed the loop after entry.

Implemented capability. Position assessments, exits, mark-to-market, portfolio Greeks, scenario monitoring, dynamic limits, risk alerts, reconciliation and continuous monitoring.

Primary inputs. Open paper positions and current market state.

Primary outputs / authority. Position health, exit instructions, reconciled marks and monitoring alerts.

Current architectural role. Closed the loop after entry.

Milestone 40 - Production API

Purpose. Decoupled UI presentation from direct database manipulation.

Implemented capability. Production API and audit surfaces for controlled programmatic/UI access.

Primary inputs. Published backend authority.

Primary outputs / authority. HTTP contracts consumed by workstation/automation.

Current architectural role. Decoupled UI presentation from direct database manipulation.

Milestone 42 - Real-time Monitoring / Operations

Purpose. Improved visibility of freshness and service failures.

Implemented capability. Realtime monitoring and operational state/health foundations.

Primary inputs. Runtime/service/feed state.

Primary outputs / authority. Health, monitoring and alert evidence.

Current architectural role. Improved visibility of freshness and service failures.

Milestone 44 - Institutional Market Structure

Purpose. Provided structure evidence beyond simple indicators.

Implemented capability. Institutional/normalized market-structure persistence and analytics.

Primary inputs. Underlying market observations.

Primary outputs / authority. Market-structure snapshots and normalized features.

Current architectural role. Provided structure evidence beyond simple indicators.

Milestone 45 - Market Overview

Purpose. Gave every symbol-level decision a market context.

Implemented capability. Database-backed market overview with market health/breadth/regime/sector/liquidity/volatility/risk context.

Primary inputs. Broad-market/sector/index observations.

Primary outputs / authority. Current market-overview snapshots.

Current architectural role. Gave every symbol-level decision a market context.

Milestone 46 - Market Intelligence

Purpose. Made run_market_ingestion.py a primary operational entry point.

Implemented capability. Authoritative market-ingestion orchestration and persistence/publication; Polygon-authoritative closure.

Primary inputs. Market/option observations.

Primary outputs / authority. Published market intelligence and ingestion publications.

Current architectural role. Made run_market_ingestion.py a primary operational entry point.

Milestone 47 - Published-State Consumption and Lineage

Purpose. Formalized “data exists” vs “this is the current certified authority.”.

Implemented capability. Published-state resolver, scanner/decision consumption, staleness/failure governance, persistent lineage, reporting, historical replay and certification.

Primary inputs. Multiple snapshots/publications.

Primary outputs / authority. Current READY authority with lineage and stale/missing semantics.

Current architectural role. Formalized “data exists” vs “this is the current certified authority.”.

Milestone 48 - SPX / NDX / RUT Index Integration

Purpose. Improved broad-market confirmation.

Implemented capability. Integrated major U.S. index instruments into authoritative market context and scanner intelligence.

Primary inputs. Index observations.

Primary outputs / authority. Index-aware market/trend context.

Current architectural role. Improved broad-market confirmation.

Milestone 49 - Authoritative Paper Trading Data Model

Purpose. Retired fragmented JSON state as authoritative paper-trading truth.

Implemented capability. Canonical orders/events, paper executions/fills, cash reservations, sessions, checkpoints, controls, marks and lifecycle with PostgreSQL authority.

Primary inputs. Canonical order intent and paper fills.

Primary outputs / authority. Transactional authoritative paper ledger.

Current architectural role. Retired fragmented JSON state as authoritative paper-trading truth.

Milestone 50 - IBKR Paper Trading Integration

Purpose. Kept market data authority separate from broker truth.

Implemented capability. Paper account binding, account/position snapshots, reconciliation and broker routing with paper-only enforcement.

Primary inputs. IBKR Paper account/order/fill truth.

Primary outputs / authority. Normalized broker snapshots and reconciled PostgreSQL state.

Current architectural role. Kept market data authority separate from broker truth.

Milestone 51 - Automated Position & Exit Lifecycle

Purpose. Reduced manual lifecycle gaps.

Implemented capability. Automated position marking, lifecycle transitions and exits around authoritative PostgreSQL state.

Primary inputs. Open positions and current market/risk state.

Primary outputs / authority. Exit instructions and lifecycle events.

Current architectural role. Reduced manual lifecycle gaps.

Milestone 52 - Trend Intelligence

Purpose. Made multi-timeframe trend a first-class upstream authority.

Implemented capability. Daily/weekly/monthly trend state, transitions, forecasting and institutional participation/leadership integrated into market ingestion.

Primary inputs. Underlying price history across timeframes.

Primary outputs / authority. Trend state, transition, forecast and participation snapshots.

Current architectural role. Made multi-timeframe trend a first-class upstream authority.

Milestone 53 - Trend/Decision UI Refinement and Dealer

Purpose. Improved usability without moving authority into the UI.

Implemented capability. Workstation refinements and correction/hardening of dealer gamma-flip intelligence.

Primary inputs. Published trend/dealer state.

Primary outputs / authority. Clearer UI presentation and corrected dealer context.

Current architectural role. Improved usability without moving authority into the UI.

Milestone 54 - Canonical Opportunity Domain

Purpose. Reduced cross-module shape drift.

Implemented capability. Canonical opportunity data model linking scanner candidates to downstream workflows.

Primary inputs. Scanner/stock intelligence candidate.

Primary outputs / authority. Stable opportunity identity and lineage.

Current architectural role. Reduced cross-module shape drift.

Milestone 55 - Institutional Intelligence

Purpose. Created a unified explanation layer.

Implemented capability. Institutional intelligence persistence, services and workstation view.

Primary inputs. Market/stock/options/decision evidence.

Primary outputs / authority. Aggregated institutional intelligence.

Current architectural role. Created a unified explanation layer.

Milestone 56 - Advanced Trade Builder

Purpose. Separated trade construction from execution.

Implemented capability. Advanced trade-plan construction for exact option legs and strategy semantics.

Primary inputs. Approved option strategy/contract set.

Primary outputs / authority. Structured trade plan ready for validation/certification.

Current architectural role. Separated trade construction from execution.

Milestone 57 - Portfolio Intelligence

Purpose. Made portfolio context visible during candidate decisions.

Implemented capability. Portfolio intelligence persistence and UI.

Primary inputs. Positions, cash, risk and allocation state.

Primary outputs / authority. Portfolio-level intelligence views.

Current architectural role. Made portfolio context visible during candidate decisions.

Milestone 58 - Performance Learning

Purpose. Prepared continuous learning while preserving governance.

Implemented capability. Performance-learning state and UI.

Primary inputs. Trades, fills, outcomes.

Primary outputs / authority. Attribution/calibration/learning evidence.

Current architectural role. Prepared continuous learning while preserving governance.

Milestone 59 - Institutional Execution Workspace

Purpose. Created a governed last-mile before routing.

Implemented capability. Execution workspace persistence and UI for executable plans.

Primary inputs. Certified/staged trade plan and current quotes.

Primary outputs / authority. Execution-intent state and operator workflow.

Current architectural role. Created a governed last-mile before routing.

Milestone 60 - Native IBKR Atomic Combo Execution

Purpose. Avoided treating spreads as unrelated single-leg orders.

Implemented capability. Native multi-leg combo representation/routing where broker supports it.

Primary inputs. Multi-leg trade plan.

Primary outputs / authority. Atomic/combo broker order semantics.

Current architectural role. Avoided treating spreads as unrelated single-leg orders.

Milestone 61 - Unified Underlying / Stock Intelligence

Purpose. Established the underlying-first source for Institutional Options.

Implemented capability. Multi-timeframe stock intelligence, scoring, structural levels/zones, dealer context, position intelligence and outcome learning.

Primary inputs. Market/trend/underlying data.

Primary outputs / authority. Current Stock Intelligence publications and scanner rows.

Current architectural role. Established the underlying-first source for Institutional Options.

Milestone 62 - Underlying-first Institutional Options

Purpose. Prevented option scanning from drifting away from the underlying thesis.

Implemented capability. Strategy generation from current Stock Intelligence, contract optimization, probability/valuation/management and downstream handoff.

Primary inputs. Current Stock Intelligence plus option chain.

Primary outputs / authority. Ranked strategy alternatives with exact contracts and plans.

Current architectural role. Prevented option scanning from drifting away from the underlying thesis.

Milestone 63 - Broker Portfolio Synchronization

Purpose. Kept internal portfolio state reconciled with broker truth.

Implemented capability. Recurring broker account/position synchronization into normalized portfolio authority.

Primary inputs. IBKR Paper account/positions.

Primary outputs / authority. Broker portfolio snapshots/publications.

Current architectural role. Kept internal portfolio state reconciled with broker truth.

Milestone 64 - Portfolio Risk & Capital Allocation /

Purpose. Ensured candidates cannot bypass portfolio risk/capital governance.

Implemented capability. Portfolio Greeks, VaR/ES, concentration/heat, portfolio-aware sizing/allocation and decision publication; M64.2 corrected defined-debit risk semantics and hardened current-authority lineage/retirement.

Primary inputs. Current positions, candidates, risk snapshots and Stock Intelligence run.

Primary outputs / authority. Risk snapshot, optimization snapshot, current portfolio decisions and allocation publication.

Current architectural role. Ensured candidates cannot bypass portfolio risk/capital governance.

Milestone 65 - Performance Command Center & Outcome

Purpose. Connected outcome evidence back to future decision governance.

Implemented capability. Prediction-vs-actual attribution, probability calibration, execution/management quality and governed continuous-learning evidence.

Primary inputs. Completed trade outcomes and forecasts.

Primary outputs / authority. Performance dashboards, attribution and calibration evidence.

Current architectural role. Connected outcome evidence back to future decision governance.

Milestone 66 - Production Operations & Reliability

Purpose. Made reliability a dedicated platform capability.

Implemented capability. Scheduler, health, dependency/freshness monitoring, recovery/replay, alerts, watchdogs/SLOs and operational governance.

Primary inputs. Service/job/data state.

Primary outputs / authority. Operational health/readiness evidence and automated recovery controls.

Current architectural role. Made reliability a dedicated platform capability.

Milestone 67 - Live Trading Governance & Certification

Purpose. Live trading remains explicitly gated, not implied by paper readiness.

Implemented capability. Environment isolation, approvals, audit, kill switch, rollback/recovery and live-trading certification controls.

Primary inputs. Execution environment and governance state.

Primary outputs / authority. Activation/deny controls and audit evidence.

Current architectural role. Live trading remains explicitly gated, not implied by paper readiness.

Milestone 68 - Governed Inflection Intelligence

Purpose. Added timing/turning-point context without bypassing downstream controls.

Implemented capability. Multi-factor/multi-timeframe directional inflection integrated into Stock Intelligence, options, valuation, decision and Trade Builder; later authority/validation hardening.

Primary inputs. Underlying/market structure/trend evidence.

Primary outputs / authority. Directional inflection score/state and governed timing evidence.

Current architectural role. Added timing/turning-point context without bypassing downstream controls.

Milestone 69 - Option Valuation & Relative Value

Purpose. Made price attractiveness distinct from directional attractiveness.

Implemented capability. Independent fair value, volatility/surface/dealer-flow/execution/event/relative-value mispricing, edge attribution/stability and multi-leg package valuation; event automation.

Primary inputs. Option quotes/surface, underlying, events, peer groups.

Primary outputs / authority. Fair value, mispricing classification and edge attribution.

Current architectural role. Made price attractiveness distinct from directional attractiveness.

Milestone 70 - Institutional Execution Intelligence

Purpose. Blocked otherwise-good trades when current execution conditions were unacceptable.

Implemented capability. Executable quote/freshness/liquidity/slippage/drift intelligence and execution completion analytics.

Primary inputs. Certified plan plus current option quotes.

Primary outputs / authority. Execution preflight and quality evidence.

Current architectural role. Blocked otherwise-good trades when current execution conditions were unacceptable.

Milestone 71 - OPEX & Futures Intelligence

Purpose. Broadened cross-market confirmation.

Implemented capability. Expiration-cycle and futures/cross-asset intelligence with governed authority and scheduled ingestion.

Primary inputs. Options expiration and futures observations.

Primary outputs / authority. OPEX/futures snapshots used as market context.

Current architectural role. Broadened cross-market confirmation.

Milestone 72 - Outcome Calibration & Execution Quality

Purpose. Kept learning dependent on trustworthy execution lifecycle.

Implemented capability. Outcome calibration/execution quality framework later parked while lifecycle prerequisites were completed.

Primary inputs. Forecasts, fills, outcomes.

Primary outputs / authority. Calibration/execution-quality evidence.

Current architectural role. Kept learning dependent on trustworthy execution lifecycle.

Milestone 73 - Autonomous Dynamic Position

Purpose. Made the platform responsible for lifecycle after approval, not just trade entry.

Implemented capability. Broker-truth lifecycle, min tick, repricing/chase controls, retry/rearm, adaptive order lifetime, cancellation reconciliation and autonomous entry/exit management.

Primary inputs. Execution intent, broker orders/quotes/fills, position state.

Primary outputs / authority. Governed working-order and autonomous management state.

Current architectural role. Made the platform responsible for lifecycle after approval, not just trade entry.

Milestone 74 - Autonomous Management Hardening

Purpose. Reduced ambiguity around whether autonomous exits are actually active.

Implemented capability. Unified strategy-level management, exit health and freshness-aware revalidation across single/multi-leg positions.

Primary inputs. Open managed positions.

Primary outputs / authority. Clear management health and robust exits.

Current architectural role. Reduced ambiguity around whether autonomous exits are actually active.

Milestone 75 - Trade Plan Certification

Purpose. Prevents stale/uncertified recommendations from becoming orders.

Implemented capability. Deterministic certification requiring a current validated plan before execution.

Primary inputs. Trade Builder plan plus current authorities.

Primary outputs / authority. Certified plan or explicit certification failure.

Current architectural role. Prevents stale/uncertified recommendations from becoming orders.

Milestone 76 - Institutional Decision Intelligence

Purpose. Prepared a stable champion baseline for M77 research.

Implemented capability. Current decision authority, publication/lineage and downstream consumption hardening.

Primary inputs. Current intelligence/risk/valuation evidence.

Primary outputs / authority. Authoritative institutional decisions.

Current architectural role. Prepared a stable champion baseline for M77 research.

Milestone 77 - Outcome Probability, Historical Research

Purpose. Build a governed outcome-probability and historical/prospective research program without allowing research to mutate production automatically.

Implemented capability. Historical underlying replay, long-history replication, multi-cadence experiments, downside-risk veto research, probability ranking, management geometry, capital-priority/capacity challengers, evidence registry and prospective frozen protocols.

Primary inputs. Frozen production Stock Intelligence/decision baseline, historical underlying authority and prospective daily snapshots.

Primary outputs / authority. Certified/rejected research evidence, production-certified DRVE downside veto, PROBABILITY_UP ranking evidence, CPRE/CACA prospective research records and research-program closure state.

Current architectural role. Historical discovery is closed. M77 now serves as the governed evidence/maturity layer for frozen prospective protocols and supplies matured barrier outcomes consumed by M78 Setup Intelligence.

Milestone 78 - Governed Institutional Setup Intelligence & Conditional Alpha Engine

Purpose. Convert setup semantics that previously existed as scattered detector features, execution geometry or broad directional labels into explicit, auditable research populations whose conditional edge can be measured prospectively.

Implemented capability. Canonical setup taxonomy/lifecycle; trend pullback/continuation; breakout/breakdown setup-confirmation-retest-continuation; failed-breakout/failed-breakdown reversals; support/resistance reversals; point-in-time PEAD research archetypes; immutable snapshots; lifecycle transitions; M77 outcome linkage; hierarchical empirical probability; shrinkage/readiness gates; expected-value/capital/time-efficiency calculations; shadow option-expression; cross-sectional ranking; challenger approval/activation; prospective certification; publications and audit.

Governance. Additive and shadow-only. No production source or authority is replaced. Automatic training, activation, certification and champion promotion are disabled. INSUFFICIENT_EVIDENCE fails closed.

Observed current state. A successful capture processed 614 candidates into 960 setup snapshots. At the observed point outcomes had not yet matured sufficiently for training, so no shadow model was active.

Post-M78 operational hardening

M77 and M78 now run through one Monday-Friday 18:30 LaunchAgent with an explicit M77 READY gate. M77.40 now derives effective market date from Stock Scanner lineage rather than portfolio publication processing timestamp. Scheduled market/event ingestion is Monday-Friday only while continuous operational services remain unchanged.